HAT-P-32 b

R-0004

Practice run · workflow W8-validation · record `2026-07-07_HAT-P-32b_sparse` · created 2026-07-08T01:22:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458107.714 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1584 +/- 0.0064
Transit depth (Rp/Rs)^2	2.51 +/- 0.2 [%]
Orbital Inclination (inc)	88.07 +/- 1.41
Airmass coefficient 1 (a1)	1.1552 +/- 0.0061
Airmass coefficient 2 (a2)	-0.1169 +/- 0.0047
Scatter in the residuals of the lightcurve fit is	0.53 %
Best Comparison Star	None
Optimal Aperture	4.19
Optimal Annulus	10.21
Transit Duration (day)	0.1301 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1584 ± 0.0064 vs 0.1489 (deviation 0.95σ)
Duration fit vs published	0.1301 d vs 0.1304 d (deviation 0.1σ)
Mid-time O–C	4.3 min
Depth SNR	15.8
Residual scatter	0.53 %

Light curve

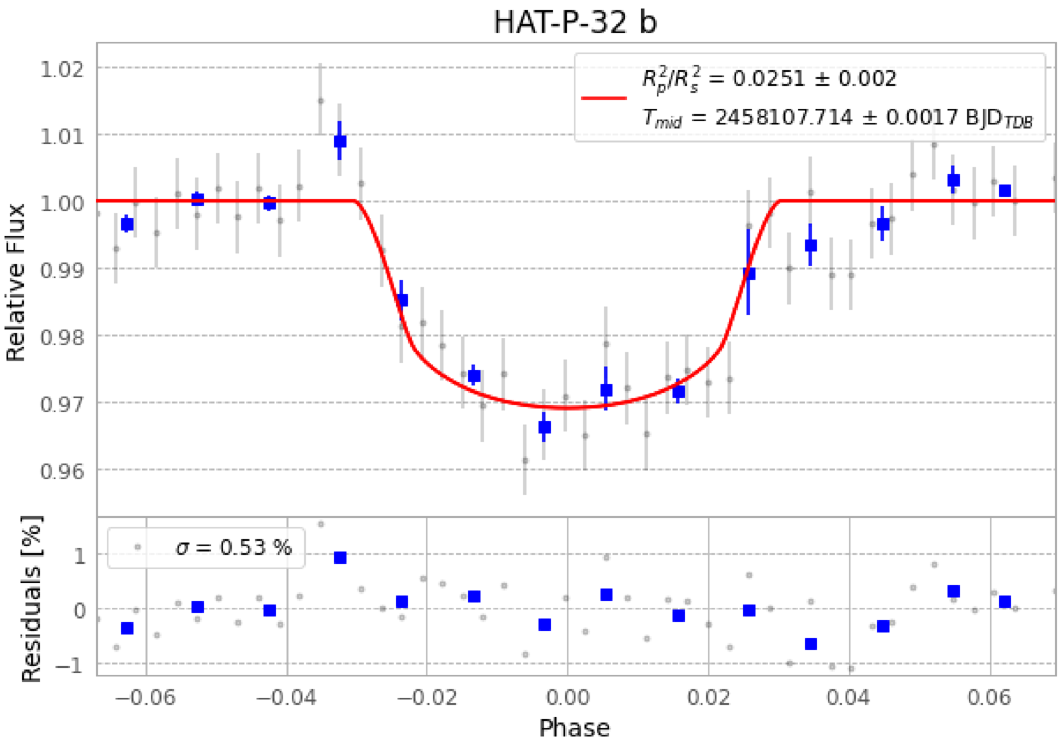

Figure 1 — FinalLightCurve_HAT-P-32 b_17-December-2017.png (SHA-256 98d15e64bbf98acf…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 286], 56 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 841d71f2aff13b84…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

48 FITS frames (32 MB), `HATP-32171220013343.FITS` → `HATP-32171220083612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: exotic-sparse.log (082528d70227…), FinalLightCurve_HAT-P-32 b_17-December-2017.png (98d15e64bbf9…), FinalLightCurve_HAT-P-32 b_17-December-2017.csv (e4f90f4f5416…)

